# Marshall Wilcox

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## Computer Skills

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|---|---|
| **Languages / Frameworks** | React, JavaScript, C#, HTML, Ember.JS, SQL |
| **Tools/Applications** | Node.JS, NPM, Yarn, Docker, Azure Kubernetes Service, Azure Container Apps, Visual Studio, VS Code, SQL Server, Internet Information Server, Microsoft Content Management Server, InstallShield Developer, nCipher Hardware Security Modules (HSMs) |
| **Methodologies** | Scrum, Agile, Pair Programming, Test Driven Development, Design Patterns |

## Work Experience

### Microsoft Corporation

**Principal Group Engineering Manager / Lead – Microsoft Commerce/Xbox** **September 2022 – July 2026**

**Purchase Experiences**

- Led a team as large as a mix of 18 leads and individual contributors building Microsoft's Purchase Experiences used by millions of customers across dozens of endpoints to purchase products and redeem codes for billions of dollars' worth of hardware, software, and subscriptions annually
- Designed and successfully delivered full rewrites of legacy purchase flows into React with server-side initial rendering, delivering massive performance improvements while modernizing our codebase and infrastructure
- Designed and released Guest Checkout purchase flows for Microsoft.com and Xbox.com, delivered Apple Pay and Google Pay payment integrations, built Buy Now, Pay Later and General Financing purchase flows, and partnered with a wide variety of partner teams across Microsoft to deliver UI integrations of our Purchase Experiences into their clients
- Delivered updates to support controller navigation and order data attribution for the Purchase Experiences in Windows 11 for gaming handhelds as well as launched new purchase flows for Xbox games streaming on supported Smart TV platforms
- Drove pre-order preparations and launch support for selling the ASUS ROG Xbox Ally gaming handhelds from Microsoft.com and Xbox.com
- Partnered with the overseas Xbox.com engineering team to build a custom financing flow for Xbox console purchases
- Delivered small business purchase flows on Microsoft.com, allowing small business customers to be able to purchase hardware, warranties, and Microsoft 365 for their small business using their business identity
- Contributed to AI driven projects including version upgrades for React and Node including migration of thousands of functional and unit tests, an Agentic Checkout prototype for Microsoft.com via the Google Universal Commerce Protocol (UCP), and hackathons for AI driven product recommendations, and AI non-partisan Political Election information
- Launched Xbox Mastercard, working with partners across Xbox and Microsoft as well as Barclays US on user flows and features required for Microsoft to offer an Xbox-branded credit card with rewards points redeemable during purchase
- Managed multiple employees into successful transitions to the Senior and Principal bands at Microsoft and into becoming Engineering Leads, ensuring they were ready for added scope and responsibility and that they delivered impact once they were at their new level

- Led team through multiple division-level restructures focusing on retention of talent, team motivation, and high productivity during times of uncertainty
- Served as interviewer and hiring manager for engineering positions, performing resume screening, initial candidate screens, and longer interviewing sessions, with a focus on understanding user requirements and systems design. Also interviewed candidates for partner teams across Microsoft including for Program and Project Manager roles.
- Managed vendor developers as well as quality assurance engineers; creating SOWs, managing contracts and billing cycles, and supporting vendor employees to deliver features through to the end of their contracts
- Served as Executive Incident Manager for Purchase Experiences, driving large cross-team incidents through troubleshooting, mitigation, resolution monitoring, and tracking and delivery of post-mortem repair items
- Ran weekly live-site review meetings, focusing on not just alerted incidents but also proactive investigation of data anomalies and trends to identify and fix issues before they were reported by end customers

### Senior / Principal Engineering Lead - Universal Store / Dynamics 365 September 2017 – September 2022

**Purchase Experiences**

- Led diverse team of engineers to support all consumer web purchase and redeem flows at Microsoft
- Led development and deployment for several business-critical codebases for Microsoft:
    - Microsoft Store and In-app purchase flows for Games and Applications across Windows, Xbox, Web, and Windows Phone
    - Token Redemption across Web, Windows, Xbox, and Windows Phone
    - Family Funding and Ask a Parent flows both on the web and in the Microsoft Family Safety apps on Android and iOS
    - Setup.office.com token redemption site specific to Office / Microsoft 365
- Delivered successful design and release for several critical new features for Microsoft Stores including:
    - PSD2 – the Payment Systems Directive 2, a mandate by EU countries to strengthen authentication for authorizing all credit card purchases starting in September 2019
    - NGP – Microsoft's Next Generation Privacy initiative to comply with the EU's GDPR legislation
    - Contextual Store – a method for 1st party partner sites like Mixer.com as well as Xbox game streaming clients to host a purchase flow so that users can purchase products and add-ons without leaving the stream they are using
    - Store Gifting – allowing Microsoft Store users to gift Games, Apps, Subscriptions, and Add-Ons to other users using their Gamertag or an email address to deliver the gift
    - Warranty Purchase – a way for Microsoft to make additional revenue by offering post-purchase warranty upsell for eligible PCs and Xbox devices
    - Office Convergence onto Universal Store – a multi-year project to move the consumer Microsoft 365 (formerly Office 365) subscription business off their bespoke subscription system and onto the same subscription engine used by other subscription businesses
- Drove planning and delivery of a new Microsoft.com purchase flow built on top of Dynamics 365 Commerce as a showcase of building on top of that product at large scale for e-commerce sites
- Worked on proactive mitigations to prevent high visibility software supply chain attacks by working with application security teams and partner teams to catalog and sandbox all third-party advertising and customer support scripts running on Microsoft.com store pages
- Led implementation of architectural updates for the Xbox Series S/X pre-orders as well as Xbox All Access and Special Edition Halo Infinite edition launches

- Drove capacity planning and bot prevention implementations to combat Xbox console resellers, where resellers were automating UIs and APIs to attempt to purchase hardware before regular customers were able to see and purchase the inventory
- Helped design and build a gated purchase flow for Xbox consoles to minimize inventory hoarding by resellers
- Monitored Surface and Xbox hardware product launches including analyzing live-site transactional data as it came through to provide updates to business and engineering leadership stakeholders
- Served as interviewer for engineering positions, with a focus on understanding user requirements and systems design. Also interviewed candidates for partner teams across Microsoft including for Program and Project Manager roles.
- Managed multiple employees through position eliminations, treating affected employees with compassion and respect and having positive exit discussions about areas where I could grow as a people manager
- Managed and retained team members during difficult large-scale reorganizations where Purchase Experiences engineering, operations, performance, and validation teams were combined and downsized
- Served as Executive Incident Manager for Purchase Experiences, driving large cross-team incidents through troubleshooting, mitigation, resolution monitoring, and tracking and delivery of post-mortem repair items

**Senior Software Engineer – Microsoft Universal Store** **January 2014 – September 2017**

**Storefront Purchase Experiences**

- Designed and built a Cart and Checkout prototype for the original converged Microsoft Store site, proving out the viability of a Single Page App (SPA) for Cart and Checkout flows for buying hardware and software through integration with new commerce APIs being developed by partner teams
- Updated Xbox purchase flows to support new markets, enable features for compliant handling of multiple language markets, and delivered features for the expansion of Xbox One consoles to sell in China with customized console registration and purchase flows
- Helped design and build the original web-blended Ember.JS single page app (SPA) purchase flows for Windows 10 across PC, Microsoft.com, and Windows Phone, supporting new features like new ways to pay, risk challenge handling, and mobile carrier integrations that had never existed on the Microsoft purchase stacks
- Served as the Security Champ for the Microsoft.com Store website, as well as Xbox.com and the Storefront Purchase Experiences. Created and updated Threat Models, attended feature security design and signoff meetings, actioned escalations from the Microsoft Security Research Center (MSRC), and maintained security compliance for our purchase flows
- Served as virtual lead for the project to unify the Windows and Xbox purchase flows, reviewing designs, coding portions of the systems, and signing off on security aspects of the project including authentication token handling in the Windows OS as well as PIN and Passcode handling for purchases
- Served as virtual lead on the team for partner feature delivery in our code bases, overseeing integrations by performing design and code reviews of solutions from partner teams for Ask a Parent flows as well as custom China payment flows
- Served as On-Call engineer (DRI) for purchase flows across many rotations, triaging and fixing and monitoring incidents, including developing and releasing a hot fix update on Xmas day for an issue affecting new Microsoft customers setting up brand new devices

**Senior / Lead Software Development Engineer – Xbox.com / Xbox Live** **May 2008 – January 2014**

**Xbox Live Content Programming and Publishing systems**

- Served as the Software Development Engineering Lead for the Dante Content Management system in Xbox. This system served as the content management and publishing system for the Xbox 360 Console Dashboard, serving manifest files (UI definition) and assets (images, videos, and applications) to millions of customers worldwide
- Designed and developed core areas of Dante, including REST web services for generating and signing console manifests that were interpreted by the Xbox 360 Dashboard
- Partnered with the Xbox Live Catalog team to integrate new Xbox Game Catalog and later Movies, TV, and Music Catalog features into Dante
- Worked with Xbox Live Service Delivery team to set up SSL for Console assets, which allowed secure delivery of assets to the console
- Developed preview workflows, where content authors could see and test future scheduled dashboard content using specially configured consoles
- Added system extensions for Flash and Lua in Dante. These runtimes enabled the Xbox Twitter application, small Lua apps such as the avatar snowglobe and quick voting apps, and the Flash player for the Internet Videos feature
- Led the team in design and development of the Internet Videos on Xbox Live feature, a feature on the Xbox Console Dashboard that showed syndicated content from MSN (Dilbert and The New Yorker video cartoons) and NBC (Nightly News, Today Show, etc)
- Updated the Dante content system for the original Kinect Hub and later the versions of the Xbox 360 dashboard that were fully designed for using Kinect with voice and motion control to navigate the dashboard
- Integrated the Bing “Image of the Day” into our content system, allowing for the Bing section on the Xbox 360 dashboard to have the same images as Bing on the web
- Served as a key contributor providing requirements for the Xbox One dashboard data service, working with senior peers working in partner teams to ensure that needed paradigms from the Xbox 360 dashboard systems were maintained and that limitations around culture handling and content reuse were fixed in the new design
- Built catalog integration into the Xbox Live Compass content system. This integration allowed the Xbox One console dashboard to be programmed to point tiles to catalog content through an integrated catalog browser
- Led the design for a content publishing system to publish Zune hub meta-data into the Zune catalog, and later to publish Xbox 360 dashboard manifests from Compass
- Led the team in the design and delivery of Xbox SmartGlass Video Activity publishing, used for annotating movies on a second screen while you were watching on the Xbox
- Managed vendor developers for multiple projects throughout my time in Xbox, creating SOWs, managing contracts and billing cycles, and supporting the vendors to deliver features through to the end of their contracts

**Lead Developer – Product Release / Secure IP Systems** **September 2002 – April 2008**

**Secure License Key**

- Led design, development, and release of the internal Secure License Key system (SLK)
- Moved license key generation and distribution systems into an incredibly secure hosting environment where critical system tasks were completely isolated from any network
- Worked with Corporate Security, Trustworthy Computing Group, and Microsoft Research to get buyoff on our approaches for securing key digital assets
- Employed two-factor authentication, symmetric and asymmetric encryption, hardware security modules, multi-threaded .Net Services, and .Net Remoting throughout the system. Our system passed an intensive security penetration test by a world-renowned external security firm
- Developed and maintained databases storing hundreds of millions of encrypted license keys; managing database processes including setup, configuration, and backup and disaster recovery systems

- Managed multiple junior FTE peers and vendor resources to deliver system improvements including rewriting core system components from C++ into C# and adding auditing, backup, reporting, and health monitoring to the system

**Windows Vista Readiness**

- Served as Development Lead for Product Release Services Readiness for Windows Vista
- Managed vendor and FTE developers to meet critical timelines for cross-group integration points between Product Release Services (PRS), Product Activation (PA) and Windows Software Protection Platform (Windows SPP)

**Codesign Launch Wave 2008 readiness project**

- Served as the Development Lead to scale and prepare the Microsoft Codesigning system for the coordinated release of Windows Server 2008, SQL Server 2008, and Visual Studio 2008
- Led optimization of the transactional database through analysis and refactoring of SQL code
- Led optimization of the file verification process to remove duplicate file hashing, improving system throughput

### Systems Engineer– Product Release Services — June 2000 – August 2002

**Product Key Systems**

- Worked with Product Release Services program management as well as Product Activation teams and MS Research teams to define and deliver approach for remediation to combat counterfeit key generators
- Designed and developed an encryption system for securely delivering product key files to manufacturers, leveraging Windows Crypto APIs via CAPICOM to encrypt and decrypt files to protect them during transport and while at rest

**Code Signing systems**

- Led project to secure the internal Microsoft code signing website from malicious attacks, by using smart card technology to create two-factor authentication for employees submitting jobs and employing hashing and signing algorithms to prevent tampering of submitted files during processing
- Developed method for getting Xbox codesigning machines off of the corporate network programmatically while they were using cryptographic keys to prevent game signing keys from being stolen during secure operations

**Product Release Systems**

- Built a centralized SQL database to import data from all 3 release labs, centralize it, export to Excel, and publish to an extranet site, giving manufacturing partners visibility into product release and shipping pipelines
- Updated Product release and shipping systems to support Dual Layer DVDs for Xbox and later products

**Electronic Document Manager (EDM)**

- Led team to develop EDM, a Web-based document creation system that allowed document administrators to create documents using a templatized web-based system for print specifications used in Microsoft's software manufacturing processes

### Release Program Manager – Product Information and Release Services — April 1999 – May 2000

**RSM Integration / Business Extensibility**

- Played several roles including developer and project manager in delivering the Release Services Manager (RSM) to internal and external Microsoft Operations users
- Developed Acceptance Test Cases, gathering user feedback through live NetMeeting teleconferences with the European Operations Center in Dublin, Ireland as well as the Asia Pacific Operations Center in Singapore
- Developed SQL Server code to process manufacturing materials requests including integrations with command line utilities and Windows COM objects
- Developed web-based Release Reports used by release labs and manufacturers worldwide to optimize manufacturing processes

**Y2K**

- Helped design and build a Y2K product SQL database and reporting site, allowing product operations teams to track Y2K issues in released Microsoft projects and feed data for pages on Microsoft.com alerting customers of the Y2K patching status of various Microsoft products
- Updated applications and firmware on multiple lab hosted servers to be Y2K compliant
- Spent New Year's Eve 1999 in a conference room at Microsoft, supporting and verifying that product release systems worked correctly before and after the clock turned over to ensure that Microsoft could still manufacture products after the start of the new Millennium

### College Intern / Contractor at Microsoft Corporation

**Web Developer** **May 1998 – April 1999**

- Web application development using XML/XSLT, ASP, MS SQL Server, and HTML
- Redesigned / re-developed Microsoft intranet site used by Product Information and Release Services organization, optimizing performance, adding personalization functionality, building an automated email release reporting system, and improving the user interface to optimize site search and employee workflows

## Education

**University of Washington** **September 1994 - June 1998**

Bachelor of Arts: Business Administration - Information Systems (IS) Concentration

## Certification

**CISSP – Certified Information Systems Security Professional** **2005 – 2008**

https://www.isc2.org/cissp/default.aspx